General Relativity
Final Examination — Answers

Conventions. Metric signature $(-,+,+,+)$; ${\Gamma^\mu}_{\nu\rho}$ is the Levi-Civita connection (symmetric in its lower indices); $\Box = \partial^\rho \partial_\rho$; ${R^\mu}_{\nu\rho\sigma}$ is defined by

$$ {R^\mu}_{\nu\rho\sigma} = \partial_\rho {\Gamma^\mu}_{\sigma\nu} - \partial_\sigma {\Gamma^\mu}_{\rho\nu} + {\Gamma^\mu}_{\rho\lambda}{\Gamma^\lambda}_{\sigma\nu} - {\Gamma^\mu}_{\sigma\lambda}{\Gamma^\lambda}_{\rho\nu}, $$

so that $R_{\mu\nu} = {R^\rho}_{\mu\rho\nu}$ as in the question.

Question 1 (2 points)

Transformation of the Riemann tensor. Under $x^\mu \to x'^\mu(x)$, ${R^\mu}_{\nu\rho\sigma}$ is a $(1,3)$ tensor:

$$ {R'^\alpha}_{\beta\gamma\delta} = \frac{\partial x'^\alpha}{\partial x^\mu}\,\frac{\partial x^\nu}{\partial x'^\beta}\,\frac{\partial x^\rho}{\partial x'^\gamma}\,\frac{\partial x^\sigma}{\partial x'^\delta}\,{R^\mu}_{\nu\rho\sigma}. $$

(The Christoffel symbols transform inhomogeneously, ${\Gamma'^\alpha}_{\beta\gamma} = \frac{\partial x'^\alpha}{\partial x^\mu}\frac{\partial x^\nu}{\partial x'^\beta}\frac{\partial x^\rho}{\partial x'^\gamma}{\Gamma^\mu}_{\nu\rho} + \frac{\partial x'^\alpha}{\partial x^\mu}\frac{\partial^2 x^\mu}{\partial x'^\beta \partial x'^\gamma}$. In R the terms containing $\partial^2 x/\partial x'\partial x'$ cancel between the derivative terms and the quadratic $\Gamma\Gamma$ terms, which leaves only the homogeneous law above.)

The Ricci tensor is a tensor. Contract the upper index of ${R'^\alpha}_{\beta\gamma\delta}$ with its third index, $\gamma = \alpha$:

$$ R'_{\beta\delta} = {R'^\alpha}_{\beta\alpha\delta} = \frac{\partial x'^\alpha}{\partial x^\mu}\,\frac{\partial x^\nu}{\partial x'^\beta}\,\frac{\partial x^\rho}{\partial x'^\alpha}\,\frac{\partial x^\sigma}{\partial x'^\delta}\,{R^\mu}_{\nu\rho\sigma}. $$

The two Jacobians carrying the summed index α combine by the chain rule:

$$ \sum_\alpha \frac{\partial x'^\alpha}{\partial x^\mu}\frac{\partial x^\rho}{\partial x'^\alpha} = \frac{\partial x^\rho}{\partial x^\mu} = \delta^\rho_\mu. $$

Therefore

$$ R'_{\beta\delta} = \frac{\partial x^\nu}{\partial x'^\beta}\,\frac{\partial x^\sigma}{\partial x'^\delta}\,\delta^\rho_\mu\,{R^\mu}_{\nu\rho\sigma} = \frac{\partial x^\nu}{\partial x'^\beta}\,\frac{\partial x^\sigma}{\partial x'^\delta}\,{R^\rho}_{\nu\rho\sigma} = \frac{\partial x^\nu}{\partial x'^\beta}\,\frac{\partial x^\sigma}{\partial x'^\delta}\,R_{\nu\sigma}. $$

This is exactly the transformation law of a covariant $(0,2)$ tensor, so $R_{\mu\nu}$ is a tensor. ■

Question 2 (3 points)

Write ${\hat\Gamma_{\mu\nu}}^\rho = {\Gamma_{\mu\nu}}^\rho + {\Delta_{\mu\nu}}^\rho$ with

$$ {\Delta_{\mu\nu}}^\rho = \frac{1}{12}\,g^{\rho\sigma}H_{\mu\nu\sigma} \equiv \frac{1}{12}H_{\mu\nu}{}^\rho. $$

Take the covariant derivative as $\hat{\nabla}_\sigma V^\rho = \partial_\sigma V^\rho + \hat{\Gamma}_{\sigma\mu}{}^\rho V^\mu$. (The argument below is unchanged if the derivative index is taken in the second slot instead, because it only uses the antisymmetry of H.)

(a) Torsion. The torsion is $T^\rho{}_{\mu\nu} = \hat{\Gamma}_{\mu\nu}{}^\rho - \hat{\Gamma}_{\nu\mu}{}^\rho$. Since $\Gamma_{\mu\nu}{}^\rho = \Gamma_{\nu\mu}{}^\rho$, only Δ survives:

$$T^\rho{}_{\mu\nu} = \Delta_{\mu\nu}{}^\rho - \Delta_{\nu\mu}{}^\rho = \frac{1}{12}\left(H_{\mu\nu}{}^\rho - H_{\nu\mu}{}^\rho\right) = \frac{1}{6}H_{\mu\nu}{}^\rho,$$

using $H_{\nu\mu}{}^\rho = -H_{\mu\nu}{}^\rho$ (total antisymmetry). Since $H_{\mu\nu\rho}$ is in general non-zero, $T \neq 0$: **the connection is not torsion-free.** The torsion vanishes only if $H = 0$.

Metric compatibility. We need $\hat{\nabla}_\sigma g_{\mu\nu} = \partial_\sigma g_{\mu\nu} - \hat{\Gamma}_{\sigma\mu}{}^\lambda g_{\lambda\nu} - \hat{\Gamma}_{\sigma\nu}{}^\lambda g_{\mu\lambda}$. The Levi-Civita connection satisfies $\nabla_\sigma g_{\mu\nu} = 0$, so

$$\hat{\nabla}_\sigma g_{\mu\nu} = -\Delta_{\sigma\mu}{}^\lambda g_{\lambda\nu} - \Delta_{\sigma\nu}{}^\lambda g_{\mu\lambda} = -\frac{1}{12}\left(H_{\sigma\mu\nu} + H_{\sigma\nu\mu}\right) = 0,$$

because $H_{\sigma\nu\mu} = -H_{\sigma\mu\nu}$. **The connection is metric compatible.**

So $\hat{\Gamma}$ is metric compatible but has totally antisymmetric torsion (this is the connection with H-flux that appears in string theory).

(b) Geodesics. A geodesic of $\hat{\Gamma}$ satisfies

$$\frac{d^2x^\mu}{d\lambda^2} + \hat{\Gamma}_{\nu\rho}{}^\mu \frac{dx^\nu}{d\lambda}\frac{dx^\rho}{d\lambda} = \frac{d^2x^\mu}{d\lambda^2} + \Gamma_{\nu\rho}{}^\mu \frac{dx^\nu}{d\lambda}\frac{dx^\rho}{d\lambda} + \frac{1}{12}H_{\nu\rho}{}^\mu \frac{dx^\nu}{d\lambda}\frac{dx^\rho}{d\lambda}.$$

The last term vanishes identically: $H_{\nu\rho}{}^\mu$ is antisymmetric in $\nu\rho$, while $\dot{x}^\nu\dot{x}^\rho$ is symmetric. So the geodesic equation reduces to the Levi-Civita one, and the geodesics (with the same affine parameter) coincide. ■

Question 3 (4 points)

(a) Singularities. In the Schwarzschild metric, $g_{tt} = -f$ and $g_{rr} = 1/f$ are singular at $r = 0$ and at $r = 2GM$. Coordinate components alone do not decide whether a singularity is physical, so compute the Kretschmann scalar $K = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$, which is coordinate-independent.

In an orthonormal frame the independent non-zero Riemann components of Schwarzschild are

$$R_{\hat{t}\hat{r}\hat{t}\hat{r}} = -\frac{2GM}{r^3}, \quad R_{\hat{t}\hat{\theta}\hat{t}\hat{\theta}} = R_{\hat{t}\hat{\phi}\hat{t}\hat{\phi}} = \frac{GM}{r^3}, \quad R_{\hat{r}\hat{\theta}\hat{r}\hat{\theta}} = R_{\hat{r}\hat{\phi}\hat{r}\hat{\phi}} = -\frac{GM}{r^3}, \quad R_{\hat{\theta}\hat{\phi}\hat{\theta}\hat{\phi}} = \frac{2GM}{r^3}.$$

Each component R_{abab} (with $a \neq b$) appears four times in the full contraction, and the signs from raising $\hat{t}$ indices square away. Summing the squares of the six independent components, with weights $4, 1, 1, 1, 1, 4$:

$$K = 4\,\frac{G^2M^2}{r^6}\left(4 + 1 + 1 + 1 + 1 + 4\right) = \frac{48\,G^2M^2}{r^6}.$$

- At $r = 2GM$: $K = \dfrac{48\,G^2M^2}{64\,G^6M^6} = \dfrac{3}{4G^4M^4}$, which is finite. So $r = 2GM$ is a *coordinate* singularity (a horizon), removable by a change of coordinates.
- At $r = 0$: $K \to \infty$. So $r = 0$ is a *genuine* curvature singularity.

You should be concerned about $r = 0$. For a large black hole, crossing $r = 2GM$ is locally unremarkable. Inside the horizon $f < 0$, so r is a timelike coordinate and r decreases along every future-directed worldline. The singularity at $r = 0$ is therefore in your future and cannot be avoided. The radial tidal acceleration is $\ddot{\xi}^r \simeq (2GM/r^3)\,\xi^r$, which grows without bound as $r \to 0$ (spaghettification).

(b) Ingoing Eddington–Finkelstein coordinates. Define the tortoise coordinate by $dr_* = dr/f$:

$$r_* = \int \frac{dr}{f} = \int \frac{r\,dr}{r - 2GM} = \int \left(1 + \frac{2GM}{r - 2GM}\right) dr = r + 2GM \ln|r - 2GM| + \text{const.}$$

Since $|r - 2GM| = 2GM\,\left|\frac{r}{2GM} - 1\right|$, the constant only shifts v, so $v = t + r_*$ with $r_* = r + 2GM \ln\left|\frac{r}{2GM} - 1\right|$, as given.

From $v = t + r_*(r)$: $dt = dv - \dfrac{dr}{f}$, so

$$dt^2 = dv^2 - \frac{2\,dv\,dr}{f} + \frac{dr^2}{f^2}.$$

Substituting into $ds^2 = -f\,dt^2 + f^{-1}dr^2 + r^2 d\Omega^2$:

$$-f\,dt^2 = -f\,dv^2 + 2\,dv\,dr - \frac{dr^2}{f}, \qquad ds^2 = -f\,dv^2 + 2\,dv\,dr + \frac{dr^2}{f} - \frac{dr^2}{f} + r^2 d\Omega^2.$$

The dr^2/f terms cancel, giving

$$\boxed{ds^2 = -\left(1 - \frac{2GM}{r}\right) dv^2 + 2\,dv\,dr + r^2\left(d\theta^2 + \sin^2\theta\,d\phi^2\right)}$$

Why this supports (a). The metric components are now regular at $r = 2GM$. The (v, r) block $\begin{pmatrix} -f & 1 \\ 1 & 0 \end{pmatrix}$ has determinant $-1 \neq 0$ for all r, and its inverse $g^{vv} = 0$, $g^{vr} = 1$, $g^{rr} = f$ is finite there. So the coordinates cover the horizon smoothly, and the apparent divergence at $r = 2GM$ in the Schwarzschild form was a feature of the t coordinate. Ingoing null rays are $v = \text{const}$, and an infalling observer crosses $r = 2GM$ at finite v with nothing singular happening. At $r = 0$, however, $f = 1 - 2GM/r \to -\infty$, so $g_{vv} \to \infty$ and the metric is still singular, consistent with $K \to \infty$. Only $r = 0$ is a true singularity, and it is the one to worry about.

Question 4 (5 points)

The de Sitter metric is $ds^2 = -dt^2 + a(t)^2\, d\Sigma_3^2$ with

$$a(t) = L\cosh(t/L), \qquad d\Sigma_3^2 = d\chi^2 + \sin^2\chi\left(d\theta^2 + \sin^2\theta\, d\phi^2\right).$$

(a) Candidate cosmology; scale factor. The scale factor is $a(t) = L\cosh(t/L)$. The metric is of Friedmann–Lemaître–Robertson–Walker form with $k = +1$ (closed spatial sections, a 3-sphere) and is spatially homogeneous and isotropic, which is the cosmological principle that describes our universe on large scales.

To check that it solves Einstein's equations, use the Friedmann equations with $\rho = 0$ and cosmological constant Λ:

$$H^2 + \frac{k}{a^2} = \frac{\Lambda}{3}, \qquad \frac{\ddot{a}}{a} = \frac{\Lambda}{3}.$$

With $\dot{a} = \sinh(t/L)$ and $H = \dot{a}/a = \tanh(t/L)/L$:

$$H^2 + \frac{1}{a^2} = \frac{\tanh^2(t/L) + \operatorname{sech}^2(t/L)}{L^2} = \frac{1}{L^2} \quad \Rightarrow \quad \Lambda = \frac{3}{L^2}.$$

Also $\ddot{a} = \frac{1}{L}\cosh(t/L)$, so $\ddot{a}/a = 1/L^2 = \Lambda/3$, consistent with the second equation.

So this is a vacuum solution with positive cosmological constant $\Lambda = 3/L^2$. It is a candidate for our universe because the positive Λ produces accelerated expansion ($\ddot{a} > 0$), which is what observations of the late universe show (dark energy). Two caveats: the spatial sections are closed ($k = +1$), while observations favour spatial flatness, and the model contains no matter, so it does not describe the matter-dominated early universe. At late times $a \simeq \frac{L}{2}e^{t/L}$, which is the exponential expansion expected in a Lambda-dominated future.

(b) Conformal rescaling and causal structure. Let $g_{\mu\nu} = \Omega^2\bar{g}_{\mu\nu}$ with Ω smooth and non-zero. A curve is null if $g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = 0$. Since $\Omega^2 \neq 0$,

$$g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = \Omega^2\,\bar{g}_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = 0 \iff \bar{g}_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = 0.$$

Also $\Omega^2 > 0$ preserves the sign of ds^2, so timelike, null and spacelike vectors are the same in both metrics, and so is the time orientation. Hence the light cones coincide, and so do the causal relations ($J^\pm$, $I^\pm$). Null geodesics are also the same curves up to reparametrisation. Therefore the two metrics share the same causal structure, even though their distances and curvatures differ.

(c) Conformal time and Penrose diagram. Let $\cosh(t/L) = \sec\eta$ with $\eta \in (-\pi/2, \pi/2)$. Since $\cosh \geq 1$ for all t, this covers all of $t \in \mathbb{R}$, with $\eta \to \pm\pi/2$ as $t \to \pm\infty$.

Then $\sinh^2(t/L) = \cosh^2(t/L) - 1 = \sec^2\eta - 1 = \tan^2\eta$, so $\sinh(t/L) = \tan\eta$ (with the same sign as t). Differentiating $\cosh(t/L) = \sec\eta$:

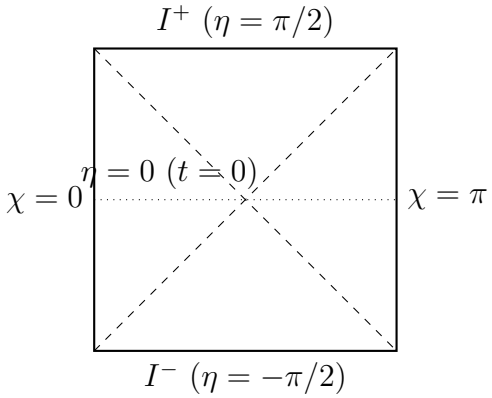
$$\frac{1}{L}\sinh(t/L)\, dt = \sec\eta\tan\eta\, d\eta \quad \Rightarrow \quad \frac{1}{L}\tan\eta\, dt = \sec\eta\tan\eta\, d\eta \quad \Rightarrow \quad dt = L\sec\eta\, d\eta.$$

Substituting into the metric, with $-dt^2 = -L^2 \sec^2\eta\, d\eta^2$ and $L^2\cosh^2(t/L) = L^2\sec^2\eta$:

$$ds^2 = \frac{L^2}{\cos^2\eta}\left(-d\eta^2 + d\chi^2 + \sin^2\chi\, d\Omega^2\right).$$

So de Sitter is conformal to the strip $-\pi/2 < \eta < \pi/2$, $0 < \chi < \pi$ times S^2, with conformal factor $\Omega = L\sec\eta$. The conformal factor is finite for $|\eta| < \pi/2$.

Penrose diagram. Null rays satisfy $d\eta = \pm d\chi$, so they are at 45°. The boundaries $\eta = \pm\pi/2$ are $I^{\pm}$ (where the areal radius $L\sin\chi\sec\eta$ diverges), and the edges $\chi = 0, \pi$ are the worldlines of the two antipodal poles. Each interior point is a 2-sphere of areal radius $L\sin\chi\sec\eta$. The result is a square:



Vertical sides: worldlines of the poles $\chi = 0, \pi$. Top and bottom: future and past infinity. Dashed diagonals: the cosmological event horizons of the pole observers. The square is drawn with equal scale on both axes, so the null lines are at 45°.

Question 5 (6 points)

(a) Gauge choice and gravitational waves. Under $h_{\mu\nu} \to h_{\mu\nu} + \partial_\mu\xi_\nu + \partial_\nu\xi_\mu$, the trace changes by $h \to h + 2\partial\cdot\xi$, so

$$\bar{h}_{\mu\nu} \to \bar{h}_{\mu\nu} + \partial_\mu\xi_\nu + \partial_\nu\xi_\mu - \eta_{\mu\nu}\,\partial\cdot\xi, \qquad \partial^\mu\bar{h}_{\mu\nu} \to \partial^\mu\bar{h}_{\mu\nu} + \Box\xi_\nu.$$

Choose ξ_ν with $\Box\xi_\nu = -\partial^\mu\bar{h}_{\mu\nu}$, which is always solvable. This gives the harmonic (Lorenz) gauge $\partial^\mu\bar{h}_{\mu\nu} = 0$.

In this gauge the extra terms in the linearised equation vanish:

$$\partial^\rho\partial_{(\mu}\bar{h}_{\nu)\rho} = \tfrac{1}{2}\left(\partial_\mu\partial^\rho\bar{h}_{\nu\rho} + \partial_\nu\partial^\rho\bar{h}_{\mu\rho}\right) = 0, \qquad \partial^\rho\partial^\sigma\bar{h}_{\rho\sigma} = 0.$$

The field equation reduces to $-\frac{1}{2}\Box\bar{h}_{\mu\nu} = 8\pi G T_{\mu\nu}$, i.e.

$$\Box\bar{h}_{\mu\nu} = -16\pi G\, T_{\mu\nu}.$$

In vacuum, $\Box\bar{h}_{\mu\nu} = (-\partial_t^2 + \nabla^2)\bar{h}_{\mu\nu} = 0$ is a wave equation with propagation speed $c = 1$. Its solutions are waves travelling at the speed of light: these are gravitational waves. The residual gauge freedom removes the remaining unphysical components, leaving two transverse polarisations (e.g. h_+ and $h_\times$).

(b) Proper distances between test particles. The metric is $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ with $h_{tt} = h_{ti} = 0$, $h_{xx} = h_+(t-z)$, $h_{yy} = -h_+(t-z)$.

The particles stay at fixed coordinates. The geodesic equation is $\ddot{x}^i = -\Gamma^i{}_{\mu\nu}\dot{x}^\mu\dot{x}^\nu$. At linear order

$$\Gamma^i{}_{tt} = \tfrac{1}{2}\eta^{ij}\,(2\partial_t h_{jt} - \partial_j h_{tt}) = 0,$$

since $h_{tt} = h_{tj} = 0$. Particles initially at rest have $\dot{x}^i = 0$, so $\ddot{x}^i = 0$ and they remain at fixed x^i. Also $g_{tt} = -1$, so t is their proper time.

Distance between the origin and $(L_x, 0, 0)$. On the slice $t = $ const (orthogonal to the world-lines since $g_{ti} = 0$), the proper length is $\Delta s = \int \sqrt{g_{xx}}\,dx$. At $z = 0$ we have $g_{xx} = 1 + h_+(t)$, which is independent of x, so

$$\Delta s = L_x\sqrt{1 + h_+(t)} \approx L_x\left(1 + \tfrac{1}{2}h_+(t)\right),$$

using $\sqrt{1+\epsilon} \approx 1 + \epsilon/2$ for $|h_+| \ll 1$.

Distance between the origin and $(0, L_y, 0)$. Here $g_{yy} = 1 - h_+(t)$, so

$$\Delta s = L_y\sqrt{1 - h_+(t)} \approx L_y\left(1 - \tfrac{1}{2}h_+(t)\right).$$

The two arms change in opposite senses, which is the signature of a transverse, traceless wave.

(c) Binary inspiral, merger and ringdown. Two masses m_1, m_2 (total M, reduced mass $\mu = m_1 m_2/M$) orbit each other at separation a with orbital angular frequency $\omega = \sqrt{GM/a^3}$ (Kepler's law). The system radiates gravitational waves at twice the orbital frequency. In the quadrupole approximation the power is

$$P = \frac{32}{5}\frac{G\mu^2 a^4\omega^6}{c^5} = \frac{32}{5}\frac{G^4\mu^2 M^3}{c^5 a^5},$$

using $\omega^6 = (GM)^3/a^9$. The orbital energy is $E = -\dfrac{GM\mu}{2a}$. Energy conservation, $dE/dt = -P$, gives

$$\frac{GM\mu}{2a^2}\frac{da}{dt} = -P \quad\Rightarrow\quad \frac{da}{dt} = -\frac{64}{5}\frac{G^3\mu M^2}{c^5 a^3}.$$

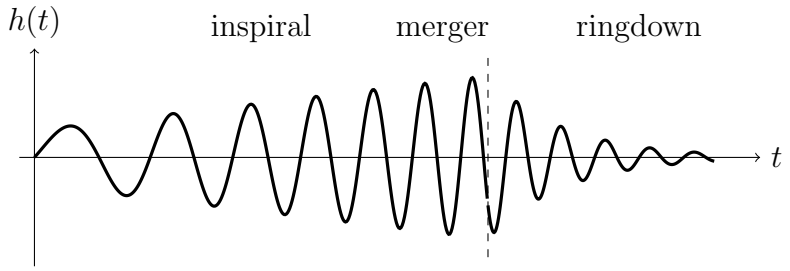
So the separation shrinks and the orbit speeds up, and the time to coalescence is $t_c = \frac{5}{256}\frac{c^5 a^4}{G^3 \mu M^2}$.

Inspiral. The orbit slowly shrinks while the frequency and amplitude of the wave increase (a "chirp"). The frequency sweeps upward, and the amplitude grows because the emitted power grows as a shrinks.

Merger. When the separation is a few times GM/c^2, the point-particle approximation breaks down and the two horizons merge into one. The emitted wave reaches its maximum amplitude and frequency here. Numerical relativity is needed to compute this stage.

Ringdown. The remnant black hole settles to a stationary state by emitting quasi-normal modes. The signal is an exponentially damped sinusoid, $h \propto e^{-t/\tau}\cos(2\pi f t)$.

Detection. A Michelson interferometer has two perpendicular arms along x and y, each of length L. A wave h_+ travelling along z changes the arm lengths by $\Delta L_x = +\frac{1}{2}h_+ L$ and $\Delta L_y = -\frac{1}{2}h_+ L$ (from part (b)). The difference $\Delta L_x - \Delta L_y = h_+ L$ shifts the phase of the light at the output port, and this phase shift is recorded. For LIGO's $L = 4$ km and $h \sim 10^{-21}$, $\Delta L \sim 10^{-18}$ m, which is a small fraction of a proton's diameter.



Sketch of the signal: the frequency and amplitude grow during inspiral, peak at merger, and ring down exponentially as the remnant settles.